

Protein Engineering Adapter via Reinforcement Learning

Failure-aware preference optimization for enzyme sequence design

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Abstract. PEARL (Protein Engineering Adapter via Reinforcement Learning) is a failure-aware protein sequence optimization framework. Its current proving ground is PETase/cutinase-family enzyme design, but the broader objective is the process: a reusable loop for moving protein language models from plausible sequence generation toward experimentally disciplined protein engineering. The project began from the natural idea that a large language model could be adapted to build protein sequences from scratch through supervised fine-tuning (SFT). That route failed as a direct design strategy: positive-only imitation learned local motifs, repeat shortcuts, and geometry proxies without reliably learning the negative constraints needed for clean, foldable, family-faithful enzyme candidates. The current PEARL program turns that failure into signal. Natural PETase/cutinase records define the positive anchor; fold-failed generated sequences and synthetic artifacts become hard negatives; and preference learning—first DPO, then sparse OPD-style teacher guidance—is used to reshape the generator distribution. The first DPO loop was small and evaluated on a microscopic slice, but it signaled a significant shift: PEARL was no longer only imitating plausible sequences; it was learning from explicit contrasts between better and worse candidates. The immediate scientific test is whether DPO and DPO+ sparse OPD reduce structural mirages while preserving novelty and enzyme-family plausibility.

1 What PEARL is Trying to Learn

Protein engineering asks a model to search an enormous sequence space without losing the physical constraints that make proteins work. Protein language models are useful because they learn statistical structure from natural sequences and can generate plausible new candidates. The hard part is that plausibility is only an entry point: a sequence can look family-like, score well under a proxy, and still fail structural or functional scrutiny. PEARL is built around that distinction. Its goal is to turn protein generation into an empirical loop: propose candidates, evaluate them through progressively stronger evidence gates, learn from the contrast between better and worse outcomes, and use that signal to steer the next generator.

The scientific thesis is simple but demanding: protein engineering systems may improve fastest when failed candidates are treated as structured learning signal rather than discarded outputs. In this framing, a bad generation is not merely waste. A high-proxy but low-confidence fold is a hard negative. A motif-preserving but globally broken candidate is a diagnostic. A repeat shortcut is evidence that the objective is under-specified.

Thesis

PEARL is moving from protein sequence generation to failure-aware protein sequence optimization. The current enzyme family is a test case; the reusable loop is the larger contribution.

and local residue patterns, but sequence plausibility is not equivalent to foldability, and foldability is not equivalent to catalytic activity. This distinction is central to PEARL.

The early PEARL campaigns repeatedly reached candidates that looked encouraging under local gates: motif presence, ESM-like proxy scores, novelty filters, or sequence-level catalytic geometry. Fold triage exposed the failure: many generated candidates remained structurally weak, repetitive, or not family-faithful under more physical evaluation. The result is not just a negative result. It defines the failure surface PEARL now optimizes against.

Modern structure prediction tools make this loop practical. AlphaFold demonstrated highly accurate structure prediction from sequence/MSA-derived inputs [5]; ESM-Fold showed that protein language models can support atomic-level structure prediction directly from sequence [6]; and ColabFold made large-scale folding workflows accessible enough to use as triage rather than a final proof of function [7]. These tools do not replace wet-lab evidence, but they make structural failure cheap enough to feed back into model training.

Protein Interpretation

Sequence plausibility $\neq$ foldability $\neq$ activity. PEARL treats these as distinct evidence layers rather than collapsing them into one score.

2 Why Protein Text is Not Enough

A protein language model learns statistical regularities over amino-acid sequences. Those regularities are useful, they encode family-like syntax, evolutionary constraints,

3 The PEARL Design Loop

PEARL is best described as a closed-loop experimental system. The loop proposes candidate sequences, passes them through evidence gates, converts the outcomes into pairwise or teacher-guided training signal, and then tests whether the updated generator improves under matched

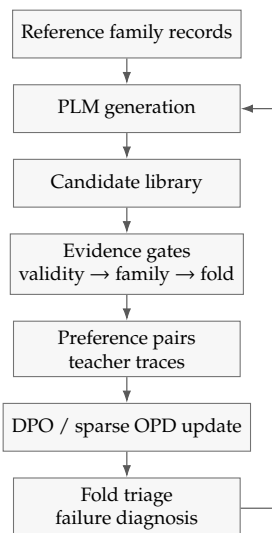


Figure 1: The PEARL loop. Failed candidates are recycled into preference data, teacher traces, and hard-negative pools rather than discarded.

evaluation.

The loop is intentionally conservative. Local sequence gates can prioritize candidates, but structural triage and future wet-lab testing remain stronger evidence. The system is designed so that failures at stronger gates flow backward into the model as rejected examples, teacher traces, or hard-negative pools.

4 Technical Vocabulary

The notation below is deliberately compact. Each equation is included beside the PEARL interpretation so readers of different technical breaths are fully aware of what is going on.

4.1 Protein sequence model

Let x be an optional design context: family prompt, scaffold, target length, mutation context, or other conditioning signal. Let $y = (a_1, \dots, a_T)$ be an amino-acid sequence. A protein language model defines a policy over sequences:

Autoregressive Sequence Likelihood

$$\pi_{\theta}(y | x) = \prod_{t=1}^T \pi_{\theta}(a_t | x, a_{<t}). \quad (1)$$

The model assigns probability to the next residue given the current context and the partial sequence. This can generate plausible protein-like strings, but the likelihood objective alone does not know which plausible strings are structurally or functionally useful.

4.2 Preference Pairs

A preference example is a triple:

Preference Pair

$$(x, y^+, y^-), \quad y^+ > y^-. \quad (2)$$

Under the same design context x , y^+ is preferred to y^- . For PEARL, the chosen side should be anchored by stronger biological evidence: natural PETase/cutinase references, vetted family-faithful candidates, or eventually wet-lab measurements. The rejected side can be a fold-failed candidate, repeat artifact, high-proxy/low-structure mirage, or synthetic negative that preserves superficial length while breaking critical plausibility.

Preference pairs encode contrast. They do not merely say “imitate this sequence.” They say: under the same design context, this candidate is better than that one, and the reason should be biologically meaningful.

4.3 DPO: Direct Preference Optimization

DPO trains the model to prefer y^+ over y^- relative to a frozen reference model π_{ref} [1]. One common objective is:

Direct Preference Optimization

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x, y^+, y^-)} [\log \sigma(\beta \Delta_{\theta})], \quad (3)$$

$$\Delta_{\theta} = \log \frac{\pi_{\theta}(y^+ | x)}{\pi_{\text{ref}}(y^+ | x)} - \log \frac{\pi_{\theta}(y^- | x)}{\pi_{\text{ref}}(y^- | x)}. \quad (4)$$

β controls how aggressively the trainable model can move away from the reference. In PEARL, DPO is attractive because it can penalize specific generated failures without first training a separate reward model or running a full PPO-style reinforcement-learning loop.

Recent work makes this direction natural rather than idiosyncratic. Rafailov et al. introduced DPO as a simpler, stable preference-optimization objective for language models [1]. ProteinDPO applied DPO to structure-conditioned protein generation aligned with experimental fitness, showing that preference optimization is already entering protein generative modeling rather than remaining only an NLP alignment tool [2].

PEARL Interpretation

DPO changes the generator distribution. Filtering only decides what we keep after sampling; DPO teaches the model that some sampled regions are worse than others.

4.4 OPD: On-Policy Distillation

OPD shifts distillation onto the student’s own trajectories. Instead of training only on static examples, the student generates sequences from its current policy, and one or more teachers provide token-level guidance on the states the student actually visits.

On-Policy Distillation

$$y \sim \pi_\theta(\cdot | x), \quad (5)$$

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{x, y \sim \pi_\theta} \sum_{t=1}^T D_t, \quad (6)$$

$$D_t = D(q_\phi(\cdot | x, y_{<t}) \| \pi_\theta(\cdot | x, y_{<t})). \quad (7)$$

q_ϕ is a teacher distribution and D is a divergence or cross-entropy-like distance. The teacher is not only grading a finished sequence; it is shaping the next-residue distribution along the path that produced the sequence.

This matters because protein design failures can be trajectory-dependent. A model may drift into a repeat, a low-complexity segment, or a family-breaking region before the final sequence is obviously bad. OPD is attractive because it can, in principle, correct the model near those states.

4.5 Multi-Teacher Consensus and Sparse OPD

Protein design is multi-objective. A candidate may improve novelty while damaging foldability, preserve a motif while breaking global structure, or look stable while leaving the enzyme family. ProteinOPD, published in May 2026, points directly in this direction: it adapts pre-trained protein language models into preference-specific teachers and distills them into a shared student using token-level OPD on the student’s own trajectories [3]. Its key idea is a normalized geometric consensus of weighted teachers.

Normalized Geometric Teacher Consensus

$$q_{\text{cons}}(a | s_t) = \frac{\prod_{k=1}^K q_k(a | s_t)^{w_k}}{\sum_{a' \in \mathcal{A}} \prod_{k=1}^K q_k(a' | s_t)^{w_k}}. \quad (8)$$

Here $s_t = (x, y_{<t})$, q_k is teacher k , and w_k is that teacher’s weight. Geometric consensus is stricter than ordinary averaging: a residue is strongly favored only when several teachers support it.

For PEARL, sparse OPD means using this idea conservatively. The system may not yet have full dense teacher logits for every token. But it can still build sparse

targets from high-confidence teacher signals; foldability, family/active-site plausibility, developability, novelty, and anti-repeat diagnostics. The objective isn’t to copy ProteinOPD; it’s to use the lesson it outlined in PEARL’s narrower enzyme-family loop.

5 Evidence Gates

PEARL should not collapse all evidence into one score. Instead, candidates are evaluated through a ladder of increasingly expensive and increasingly biological gates.

This ladder is important for both audiences. For ML readers, it prevents reward hacking from being hidden under one scalar. For protein readers, it makes clear that PEARL is not treating language-model likelihood, proxy score, or fold confidence as biochemical truth. They are staged evidence, not proof of function.

A useful shorthand is:

$$\begin{aligned} \text{validity} &\rightarrow \text{family} \rightarrow \text{novelty} \\ &\rightarrow \text{proxy} \rightarrow \text{structure} \rightarrow \text{experiment}. \end{aligned} \quad (9)$$

The closer a signal gets to the right side of the chain, the more authority it should have over training data construction.

6 Experimental Campaigns

The older language of “phases” undersells the science. The rebuilt scientific story should instead use campaigns: each campaign asked a scientific question, produced an observation, and changed the design loop.

The SFT campaign was a failure as a direct route to robust de novo enzyme design, but it was a useful failure. It showed that positive-only imitation could find surface-level bridge/repeat shortcuts while failing to learn negative topological constraints. The model could surf repeat boundaries and local geometry proxies, but it did not reliably internalize the clean single-domain constraints that matter for foldable enzyme candidates.

The manifold and proxy-guided campaigns extended this lesson. PEARL could construct and score large candidate pools, recover narrow basins, and pass local gates under some conditions. But those successes did not generalize into durable family-faithful transfer. Fold triage then turned the weak generated panel into evidence: generated rows should not be chosen positives. They are hard negatives and diagnostics.

The DPO campaign marks the first strong procedural shift. The current active dataset contains 10,000 preference pairs: reviewed natural PETase/cutinase records on the chosen side, and fold-failed generated or synthetic artifact rows on the rejected side. A 3,000-pair paid pilot

Gate	Question	Typical signal
Validity	Is this a legal protein sequence?	Amino-acid alphabet, length range, stop/invalid checks.
Degeneracy	Is it obviously collapsed?	Repeats, low-complexity segments, trivial motif padding.
Family plausibility	Does it remain in the target enzyme family?	Similarity, motifs, profile behavior, family classifier.
Novelty	Is it nontrivial?	Distance from references and training records.
Proxy utility	Does it score well computationally?	ESM/proxy/reward score, local trainability gate.
Structural triage	Does it plausibly fold?	pLDDT, PAE, fold confidence, active-site geometry, repeat artifacts.
Experimental readiness	Is it worth wet-lab spend?	Synthesis, expression, developability, assay prioritization.

Table 1: The PEARL evidence ladder. A candidate that passes an early gate can still fail a later, more physical gate.

trained for one epoch. The training objective moved in the desired direction: the first 10-batch mean DPO loss was 0.703, the last 10-batch mean loss was 0.323, and the reward margin moved from -0.0087 to 3.43 .

Small but Meaningful

The first DPO result does not prove enzyme design success. It does show that PEARL can move from imitation toward explicit contrastive learning over biological failure modes.

7 The Proxy-to-Structure Gap

The central PEARL finding so far is not that protein generation is impossible. It is that local plausibility can be misleading. A candidate can preserve a motif, pass a proxy score, and look nontrivial under sequence filters while still failing structural confidence or catalytic geometry in a folded model.

The May 30 DPO readout illustrates the current boundary. In one post-DPO slice— $p12$, temperature 0.8 , seed 7 —all 12 candidates had some serine motif, 2 of 12 had a PETase/cutinase-family motif, 2 of 12 passed sequence-level catalytic geometry, 3 of 12 passed the ESM gate, and 3 of 12 passed the local soft trainability gate. But there were 0 functional bridge hits and 0 family-faithful bridge hits. Five selected candidates were folded; their mean pLDDT range was 25.61 – 36.27 , and 0 of 5 passed CA-triad geometry.

That is a warning, not a verdict against DPO. The slice is too small to estimate yield or locate the dominant failure. But it is enough to preserve the working hypothesis: PEARL’s near-term gains are more likely to come from better negative construction, structural failure mining, and teacher-guided correction than from simply generating more candidates.

Failure Surface

PEARL has found the cliff edge where sequence-level plausibility stops being structurally credible. The next loop trains against that boundary.

8 The SFT-to-DPO Transition

SFT plainly put, failed as a design strategy. It was also the natural first attempt: adapt an LLM-like generator toward protein construction by showing it positive examples. That failure clarified the missing ingredient. Positive-only SFT can teach a model what good-looking sequences resemble; it does not automatically teach the model which nearby shortcuts are forbidden.

DPO changes the learning geometry. Instead of only imitating positives, the model sees explicit contrasts between preferred and rejected candidates. In PEARL, those rejections are not arbitrary low-scoring strings. The goal is to make them biologically meaningful: fold-failed artifacts, repeat shortcuts, high-proxy/low-fold mirages, and family-drifted candidates. This makes DPO a bridge from the failed SFT idea to a more realistic protein-engineering loop.

The optimism is procedural, not triumphalist. The first DPO loop was small, but it proved the custom-loss path works and gave a diagnostic generator to compare against sparse OPD. That is a real shift from the SFT era.

9 Hard Negatives and Sparse OPD

The current PEARL method consists of three layers. First, construct preference pairs whose chosen side is anchored by natural or strongly validated family evidence, not generated wishful positives. Second, characterize DPO-only as an active baseline across matched prompts, temperatures, seeds, and fold budgets. Third, prepare sparse OPD as a comparison branch that uses teacher consensus on student rollouts to target the structural hallucination

failure mode.

The teacher set should start small enough to diagnose: foldability/anti-hallucination, PETase-family/active-site geometry, developability, and novelty/diversity. The first sparse OPD run should not be treated as a production alignment method. It should be treated as an experiment: does teacher consensus reduce the exact failure pattern that DPO-only still shows under structural triage?

g-DPO is also relevant here because preference pair construction does not scale innocently. The number of possible pairs can grow quadratically with labeled sequence count, so clustering, redundancy pruning, and amortized likelihood computations may become necessary once PEARL moves beyond small diagnostic datasets [4].

10 Near-term Experimental Program

The next experiment should be a controlled comparison, not a broad generation sprint. Use the same prompts, seeds, temperatures, sample counts, selection rules, and fold budget across base PLM, SFT control if retained, DPO-only, sparse OPD-only if feasible, and DPO+ sparse OPD. The core readout should be structural failure reduction under novelty and family-plausibility constraints.

Hypothesis A. DPO improves candidate quality only when the rejected side includes biologically meaningful hard negatives. Test by comparing natural-positive/random-negative pairs against natural-positive/fold-failed hard-negative pairs.

Hypothesis B. Sparse OPD reduces structural mirages by correcting the student’s own generated trajectories. Test by comparing DPO-only and DPO+ sparse OPD on matched folds and repeat diagnostics.

Hypothesis C. The highest near-term leverage is better negative construction, not larger sample volume. Test by enriching the rejected pool with folded failures, low-pLDDT candidates, repeat artifacts, and high-proxy/low-structure cases, then measuring whether the same evaluation budget yields fewer structural mirages.

Minimum Viable Comparison

Base PLM vs DPO-only vs DPO+ sparse OPD; same prompts, same seeds, same temperatures, same sample counts, same selection rule, same fold budget.

11 Discipline and Limitations

PEARL should not yet be read as a validated enzyme-discovery engine. It has not demonstrated wet-lab improvement in PETase activity, and fold confidence is still only a triage signal rather than proof of function. At the same time, the first DPO loop is a meaningful positive re-

sult: even on a narrow slice, it showed a clear shift relative to the parity SFT run and suggested that preference-shaped feedback can move the generator more effectively than imitation alone. The present limitation is therefore not the absence of improvement, but the scale of evidence. The DPO signal needs to be tested across larger prompt sets, random seeds, sampling temperatures, candidate-selection rules, fold panels, and eventually experimental assays.

What PEARL has shown is still meaningful. The implementation has working generation, scoring, preference-data, DPO, sparse-target, and structural-triage components. The SFT failure is well diagnosed. The DPO path is operational. The current structural failure surface is specific enough to train against. The project has moved from a broad hope; building a protein from scratch using an LLM, to a more disciplined hypothesis: explicit preference and teacher-guided feedback may teach the generator which plausible regions of sequence space are physically untrustworthy.

Status

Established: SFT-only was insufficient; DPO infrastructure works; the proxy-to-structure gap is real in the observed panels.

Preliminary: the small DPO loop changed the training signal.

Hypothesized: sparse OPD will reduce structural mirages.

12 Conclusion

The current contribution of PEARL is a process for turning protein generation into an iterative experimental system. PETase/cutinase-family design is the proving ground because it is concrete, biologically meaningful, and difficult enough to reveal whether the loop is learning useful constraints. The broader aim is a reusable protein-engineering workflow: propose sequences, test them through progressively stronger evidence gates, convert the results into preference signal, and use each round to move the generator toward candidates that are not only plausible as sequences, but more credible as proteins.

The next scientific milestone is narrow but critical; show whether DPO and DPO+ sparse OPD reduce the modes of structural failure while preserving novelty and the plausibility of the enzyme-family under a matched evaluation. Assuming success, PEARL earns the right to scale. If it fails, the failure should still be useful, because it will identify which evidence gate, teacher, or negative-construction strategy is not yet strong enough.

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